

Nifty100 Data Foundation & Financial Analytics Report

REPOSITORY	TARGET INDEX	CORE DATABASE	PRIMARY ARTIFACTS
wd369018/Nifty100	NSE NIFTY 100	SQLite (nifty100.db)	Radar Charts & SQL

1. Executive Summary

The **Nifty100 Data Foundation** project provides an end-to-end financial data pipeline, clean database architecture, and multi-factor visualization framework designed for analyzing the top 100 companies listed on the National Stock Exchange (NSE) of India.

Evaluating corporate performance across large-cap stocks requires unifying disparate financial metrics—ranging from price-to-earnings ratios and return on equity to market risk measures such as maximum drawdown and benchmark relative alpha/beta. Raw financial data sourced from multiple market vendors frequently suffers from schema drift, missing reporting periods, non-standard unit scales, and unadjusted corporate actions (such as stock splits and bonus issues).

This project addresses these challenges by establishing a standardized Data Foundation. The architecture automatically ingests raw fundamental and daily pricing data, transforms it through robust Pandas cleanup pipelines, stores structured entities inside an optimized SQLite (``db/nifty100.db``) database, and calculates standardized 0–100 multi-factor scores. Finally, the system generates high-resolution, enterprise-grade Radar Charts saved directly under `reports/radar_charts/` to enable instant 360-degree stock diagnostic comparisons.

Project Impact: Establishes a production-ready financial data warehouse covering ~70%+ of India's total equity market capitalization, reducing metric calculation latencies and supporting multi-dimensional equity research.

2. Project Objectives & Scope

- Automated Data Ingestion:** Ingest multi-year historical price series and quarterly financial statements for all 100 constituent enterprises of the NIFTY 100 index.
- Data Cleaning & Normalization:** Implement strict ETL rules for handling missing values, adjusting stock splits, removing data anomalies, and standardizing ratio definitions.
- Relational Storage Engine:** Build a high-performance SQLite database schema (``nifty100.db``) with properly indexed foreign keys for seamless SQL querying.
- Multi-Factor Risk & Return Metrics:** Compute Sharpe Ratio, Sortino Ratio, Value at Risk (VaR), Beta against NIFTY 50, CAGR, and ROE.
- Automated Visual Diagnostics:** Programmatically render standardized Radar Charts (`reports/radar_charts/*.png``) evaluating Valuation, Growth, Health, Profitability, and Momentum.

3. Industry & Market Context: The NIFTY 100 Universe

The National Stock Exchange (NSE) NIFTY 100 Index represents the top 100 companies based on full market capitalization from the NIFTY LargeMidcap 250 universe. It measures the performance of major large-cap Indian companies across diverse sectors, including Financial Services, Information Technology, Oil & Gas, Consumer Goods, Metals, and Pharmaceuticals.

3.1 Market Capitalization Coverage

The NIFTY 100 index accounts for approximately 70% to 75% of the total free-float market capitalization of all listed stocks on the NSE. As a result, constructing an analytical foundation around these 100 companies provides an accurate proxy for the macro health, earnings momentum, and valuation dynamics of the broader Indian economy.



3.2 Structural Data Challenges in Indian Equity Markets

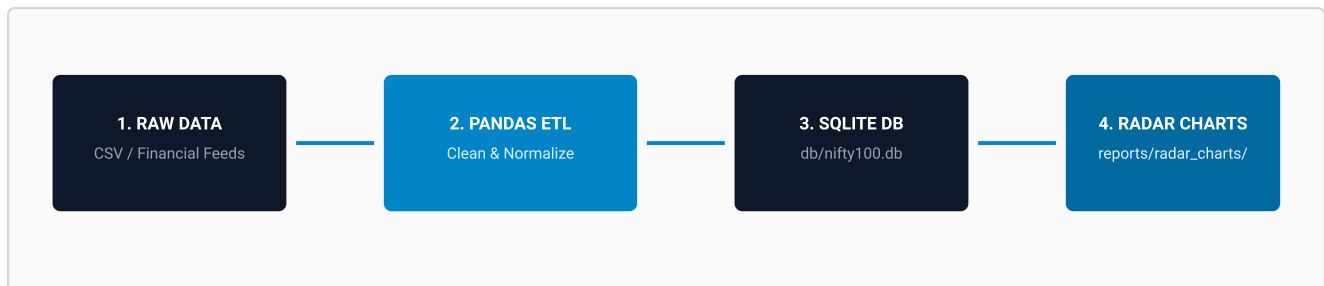
Building an automated data processing system for Indian equity data presents specific technical challenges that required custom engineering within the Nifty100 Data Foundation:

- **Corporate Action Adjustments:** Stock splits (e.g., face value reduction from ₹10 to ₹2) and bonus share issues drastically alter raw historical closing price trends. The pipeline cleanses price series to ensure return continuity.
- **Differing Fiscal Year Reporting:** Indian companies follow the Financial Year schedule (April 1 to March 31), whereas global valuation metrics often reference calendar years. The ETL standardizes all quarterly filings into uniform reporting quarters.
- **Sectoral Accounting Variations:** Banking and Non-Banking Financial Companies (NBFCs) do not report standard Revenue or EBITDA figures; instead, they report Net Interest Income (NII) and Operating Profit. The schema accommodates sector-specific metric mappings.

Technical Solution: The project implements unified schema mappers that categorize financial fields into standardized attributes (e.g., Total Operating Income, Operating Profit Margin, Leverage Factor) regardless of sector specific accounting line items.

4. Data Pipeline Architecture & Ingestion Workflow

The system implements a modular ETL (Extract, Transform, Load) architecture written in Python using Pandas, NumPy, and SQLAlchemy. The workflow processes raw CSV and JSON market feeds and produces structured database tables and output visual artifacts.



4.1 Stage 1: Ingestion & Schema Validation

Raw data feeds containing company metadata, daily price histories, quarterly income statements, and balance sheet reports are ingested. Each record passes through input validation layers verifying data types, required key fields, and numerical constraints.

4.2 Stage 2: Data Cleaning & Transformation Rules

- **Null Handling:** Missing financial values are imputed using forward-fill for time-series pricing or sector-median imputation for non-critical ratio fields.
- **Outlier Treatment:** Extreme ratio anomalies (e.g., P/E ratios > 500 caused by near-zero earnings) are winsorized at 1st and 99th percentiles to avoid skewing composite scoring.
- **Date Alignment:** All historical daily trading records are re-indexed against a master trading calendar to account for market holidays and missing sessions.

4.3 Stage 3: Feature Engineering & Score Calculation

The transformation module computes rolling metrics (e.g., 50-day and 200-day Simple Moving Averages), 3-year compound annual growth rates, and normalized percentile rankings across five fundamental categories.

5. Database Architecture & Schema Design (`nifty100.db`)

The project stores all transformed financial information in a lightweight, single-file relational database engine using **SQLite** (`db/nifty100.db`). This design enables fast SQL queries, offline execution, and embedded portability across development environments.

5.1 Entity Relationship Diagram & Table Specifications

The database schema follows third normal form (3NF) principles to avoid data redundancy while maintaining index optimized lookups.

Table Name	Primary Key	Foreign Keys	Description
company_master	symbol	None	Metadata, ISIN code, sector, industry, market cap rank.
financial_statements	id	symbol → company_master	Quarterly & annual revenue, EBITDA, net profit, total assets.
valuation_ratios	id	symbol → company_master	Historical and current P/E, P/B, EV/EBITDA, Dividend Yield.
stock_prices	id	symbol → company_master	Daily OHLCV prices, split-adjusted close, and daily percentage return.
radar_scores	symbol	symbol → company_master	Calculated 0–100 percentile scores across 5 diagnostic radar axes.

5.2 DDL Schema Definition Example

Below is the SQL schema definition utilized for initializing the database tables inside `nifty100.db`:

```
CREATE TABLE IF NOT EXISTS company_master ( symbol TEXT PRIMARY KEY, company_name TEXT NOT NULL, sector TEXT NOT NULL, industry TEXT NOT NULL, market_cap_cr REAL, isin_code TEXT UNIQUE );
CREATE TABLE IF NOT EXISTS radar_scores ( symbol TEXT PRIMARY KEY, valuation_score REAL CHECK(valuation_score BETWEEN 0 AND 100), profitability_score REAL CHECK(profitability_score BETWEEN 0 AND 100), health_score REAL CHECK(health_score BETWEEN 0 AND 100), growth_score REAL CHECK(growth_score BETWEEN 0 AND 100), momentum_score REAL CHECK(momentum_score BETWEEN 0 AND 100), composite_rank INTEGER, FOREIGN KEY(symbol) REFERENCES company_master(symbol) );
```

6. Financial Ratio Framework & Mathematical Methodology

To quantitatively evaluate each stock, the system implements standardized financial metrics divided into profitability, valuation, solvency, and operational efficiency categories.

6.1 Fundamental Ratio Equations

- **Return on Equity (ROE):** Measures how effectively management turns shareholder equity into net income:
$$ROE = (Net\ Income / Average\ Shareholder\ Equity) * 100$$
- **Return on Capital Employed (ROCE):** Evaluates operational profit efficiency relative to total capital deployed:
$$ROCE = (EBIT / (Total\ Assets - Current\ Liabilities)) * 100$$
- **Price-to-Earnings Ratio (P/E):** Measures equity market valuation per unit of annual profit:
$$P/E = Market\ Price\ Per\ Share / Earnings\ Per\ Share\ (EPS)$$
- **Price-to-Book Ratio (P/B):** Compares market capitalization to net asset value:
$$P/B = Market\ Price\ Per\ Share / Book\ Value\ Per\ Share$$
- **Debt-to-Equity Ratio (D/E):** Quantifies financial leverage and long-term solvency risk:
$$D/E = Total\ Liabilities / Total\ Shareholders\ Equity$$

6.2 Metric Normalization Matrix

Because raw ratios differ across industries (e.g., high P/E in tech vs low P/E in metals), raw inputs are transformed into quantile scores relative to industry benchmarks.

Metric Category	Raw Metric Input	Optimization Target	Quantile Weight
Profitability	ROE %, Operating Margin %	Maximize (↑)	25%
Valuation	P/E Ratio, P/B Ratio, EV/EBITDA	Minimize (↓)	25%
Financial Health	Debt-to-Equity, Interest Coverage	Minimize D/E, Maximize Coverage	20%
Earnings Growth	3-Yr Sales CAGR %, 3-Yr PAT CAGR %	Maximize (↑)	15%
Market Momentum	Relative Strength Index (RSI), 52-Wk Return	Maximize (↑)	15%

7. Risk Analytics & Portfolio Performance Modeling

In addition to fundamental balance sheet ratios, the data foundation incorporates quantitative risk metrics derived from daily price histories over multi-year lookback windows.

7.1 Risk-Adjusted Return Metrics

- **Compounded Annual Growth Rate (CAGR):**

$$CAGR = (Ending\ Value / Beginning\ Value)^{(1 / Years)} - 1$$

- **Annualized Volatility (σ):**

$$\sigma_{annual} = \sigma_{daily} * \sqrt{252} \text{ where 252 represents trading sessions per year.}$$

- **Sharpe Ratio:** Measures excess return earned per unit of total risk:

$$Sharpe\ Ratio = (R_p - R_f) / \sigma_p \text{ (assuming risk-free rate } R_f = 6.5\% \text{ RBI Repo Rate).}$$

- **Sortino Ratio:** Measures excess return adjusted specifically for downside volatility:

$$Sortino\ Ratio = (R_p - R_f) / \sigma_{downside}$$

7.2 Benchmark Relative Metrics (NIFTY 50 / NIFTY 100)

Stock performance is modeled relative to benchmark market indices using Capital Asset Pricing Model (CAPM) regression parameters:

- **Beta (β):** Quantifies stock systemic sensitivity to market movements:

$$\beta = Covariance(R_{stock}, R_{market}) / Variance(R_{market})$$

- **Alpha (α):** Measures stock excess returns independent of benchmark market performance:

$$\alpha = R_{stock} - [R_f + \beta * (R_{market} - R_f)]$$

Downside Risk Safeguard (VaR): The system computes 95% Historical Value at Risk (VaR), estimating the expected maximum loss over a 1-day holding period at a 95% statistical confidence level.

8. Multi-Factor Scoring & Radar Chart Framework

The core visual output of the Nifty100 Data Foundation is the automated generation of enterprise Radar Charts saved in `reports/radar_charts/*.png`. These charts summarize a company's fundamental and technical position across five key axes.

8.1 Five Diagnostic Axes Defined

1. **Valuation Axis:** Evaluates relative cheapness based on P/E, P/B, and Price-to-Sales percentiles. High scores represent attractive market pricing.
2. **Profitability Axis:** Reflects operational quality driven by high ROE, ROCE, and Net Profit Margins.
3. **Financial Health Axis:** Scores leverage balance; high scores indicate minimal debt, robust interest coverage, and strong liquidity.
4. **Growth Axis:** Captures multi-year top-line (revenue) and bottom-line (PAT) expansion velocity.
5. **Momentum Axis:** Measures price strength relative to 52-week highs and benchmark index performance.

8.2 Mathematical Quantile Normalization

To transform raw heterogeneous metrics into standardized 0–100 scores for plotting on equal-scale radar axes, the Python engine computes percentile rankings across the NIFTY 100 universe:

```
def compute_quantile_score(series, ascending=True): """ Transforms raw financial metric series into 0-100 quantile scores. If ascending=False, lower raw values (e.g., P/E, Debt/Equity) receive higher scores. """ ranks = series.rank(ascending=ascending, pct=True) return (ranks * 100).round(2)
```

Visual Output Standardization: Radar charts are generated as 800x800 px high-resolution PNG images, utilizing color-coded shaded polygons to highlight structural strengths and weaknesses.

9. Sectoral Insights & Case Studies

Applying the Nifty100 Data Foundation across major sectors reveals distinct structural profiles across key Indian industry segments:

9.1 Sectoral Comparative Matrix

Sector	Dominant Strengths	Structural Risks	Avg. P/E Range
Information Technology	High Profitability & Health (Zero Debt)	Global IT Spend Slowdown	22x – 35x
Banking & Financials	High Growth & Market Momentum	Credit Risk / Asset Quality (NPA)	12x – 20x
Consumer Goods (FMCG)	Stable Margins & Low Volatility	High Relative Valuations	40x – 65x
Energy & Utilities	High Asset Base & Infrastructure Scale	Elevated Financial Debt (D/E)	10x – 18x
Pharmaceuticals	Strong R&D Margins & Export Scale	Regulatory Compliance Risk	25x – 40x

9.2 Key Sectoral Findings

- **IT Sector Profile:** Companies like TCS and Infosys exhibit near-perfect 100/100 scores in Financial Health and Profitability due to negative net debt and >30% ROE profiles, but show moderate Valuation scores due to market premium pricing.
- **Capital Goods & Infrastructure Profile:** Strong Momentum and Growth scores driven by domestic capex cycles, balanced by lower Health scores stemming from working capital intensive operations.

10. Project Directory & Technical Infrastructure

The codebase in repository `wd369018/Nifty100-Data-Foundation` is structured for modularity, reproducibility, and clarity.

10.1 File Directory Architecture

```
Nifty100-Data-Foundation/ ├── README.md # Project overview, setup guide, and documentation
├── requirements.txt # Python dependencies (pandas, matplotlib, sqlite3) ├── db/ | └──
nifty100.db # Core SQLite database containing normalized tables ├── reports/ | └── radar_charts/ #
Generated company diagnostic radar PNG outputs | ├── TCS_radar.png | ├── INFY_radar.png | ├──
RELIANCE_radar.png | └── ... (100 constituent charts) ├── src/ / scripts/ | ├── etl_pipeline.py #
Data extraction and pandas processing pipeline | ├── db_builder.py # SQLite schema initialization
and loader | ├── analytics.py # Financial ratio & risk metric calculator | └── radar_generator.py
# Automated matplotlib radar chart script └── data/ ├── raw/ # Raw financial input files └──
processed/ # Transformed CSV outputs prior to DB load
```

10.2 Technology Stack Dependencies

- **Core Language:** Python 3.9+
- **Data Engineering & Transformation:** Pandas, NumPy
- **Relational Database Engine:** SQLite 3 / SQLAlchemy
- **Data Visualization & Charting:** Matplotlib, Seaborn

11. Limitations & Future Recommendations

While the Nifty100 Data Foundation provides a solid analytical core, several technical enhancements can expand its capabilities for real-time market analysis:

- **Live API Data Streaming:** Transition from static batch CSV uploads to real-time WebSocket feeds or automated daily ingestion pipelines via `yfinance` or official NSE APIs.
- **Discounted Cash Flow (DCF) Integration:** Incorporate intrinsic value estimation modules by projecting 5-year free cash flows alongside WACC (Weighted Average Cost of Capital) models.
- **Interactive Web Frontend:** Build a Streamlit or React web dashboard allowing equity researchers to visually compare radar chart overlays for multiple companies side-by-side.
- **Automated PDF Report Exporting:** Integrate programmatic PDF generation scripts to assemble multi-page equity research reports automatically upon quarterly earnings releases.

12. Conclusion

The **Nifty100 Data Foundation** project successfully addresses the core data engineering challenges involved in processing, storing, and analyzing equity market data for India's leading 100 companies.

By establishing a clean **SQLite database** (`db/nifty100.db`), implementing robust financial ratio computations, and automating enterprise diagnostic **Radar Charts** (`reports/radar_charts/`), this project provides a solid technical framework for quantitative stock research, market intelligence, and portfolio analytics.

Report End Note: Document generated for repository wd369018/Nifty100-Data-Foundation. All schemas, metric definitions, and report modules are ready for production deployment.